Math-265-A: Final Portfolio

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1 Essays

1.1 Introduction

Throughout this semester, I believe I have done a sufficient job of fulfilling the course expectations outlined in the syllabus. Throughout the reflection and portfolio, I will argue for the grade of an A in this course and will provide my reasoning with evidence to back it up. I hope you take my argument into consideration when determining my final grade for this course.

1.2 Class Citizenship

I fully believe that I have done an adequate job of being an active class participant so far, and I have done my best to contribute to class discussion in order to advance the learning of the class as a whole. Whether it be during my presentations or while I am in the audience, I always try to draw connections to other material learned in the course, and I am not afraid to present things that are wrong. I really enjoy the people who sit at my table in class (Sterling, Ezra, Matthew) because we feed off each other's energy and are always bouncing ideas off of each other in order to get a better understanding of the material. I also am not afraid to speak up in class to make corrections, ask questions, or help or encourage the people who are presenting. This class is built on the idea that we learn better as a collective, and maintaining a positive classroom environment is key to the success of the group as a whole. I have consistently made an effort to make it to class every day I am able to. The only absences I have in the class are because of travel days for gymnastics competitions, and the one time I missed class because of the stomach flu.

Throughout this entire semester, I have continued to complete assigned work for this class in a respectable manner and to my fullest ability while handing in homework and prep assignments on time. I think it is evident from how often I present in class that I am willing to put myself and my ideas out in front of the class, even if they are not 100 percent correct.

Often, on homework assignments, I have been one of the very few people to volunteer to present on certain problems that I might not be fully confident in. There has been times I might have been entirely correct or plain out wrong, but all of those times I was able to maintain my composure and facilitate class discussion about the problem at hand. Also, I am always the notetaker for the reading days I am present for, and I believe that I create discussion questions useful to understand concepts covered in the chapter. I also have posted my findings on the Q and A tab with material that has helped me understand some of the harder conceptual material in this course.

I would say I am a fairly active member of class discussion and that is represented by the amount of cards that I have accumulated since the start of presentations in this course (I would also like to note there have been a handful of days where the survey was not working so there are definitely cards missing on this graph):

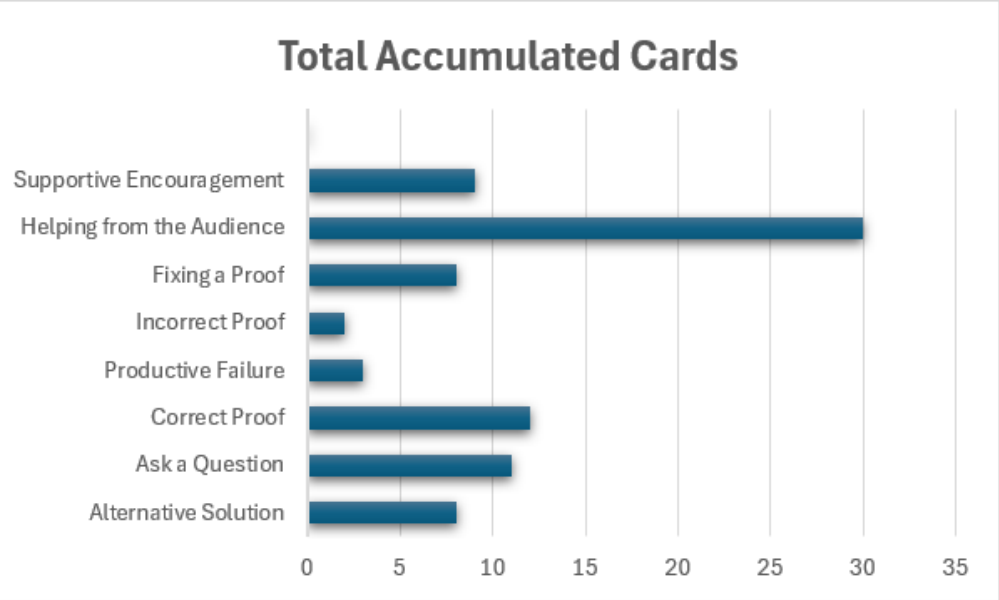


Figure 1: Total cards accumulated over the course of the semester.

1.3 Mastery of Key Mathematical Ideas and Techniques

In order to earn an A in this course, it is imperative that I have a deep understanding of the mathematical ideas and concepts that we have learned so far. In this section, I will explain each of the topics listed in the syllabus as well as include examples from the artifacts section in order to further reinforce the argument I have made for my grade.

The listed mathematical topics and techniques that we have discussed so far are:

- Truth Tables
- Translating Sentences into Symbolic Form

- Direct Proof
- Proof by Contradiction
- Proof by Contrapositive
- Proof using cases
- Proof of existence and uniqueness
- Proof by induction
- Equivalence relations
- Functions, including one-to-one and onto (injective and surjective) functions.

Truth Tables

Truth Tables are valuable tools that help us analyze the outputs of constructed mathematical statements using connectives based on the different combinations of values we have for each of the truth values for each part of the statement. The connectives that act on each part of the statement determine the output of the statement. By analyzing the input values and their corresponding outputs, we can make the argument that two statements are equivalent if their outputs are the same. This implies that two different mathematical statements can be equivalent if they have the same outputs of their corresponding truth tables. This is the foundation for determining the equivalence laws that we use in order to manipulate expressions into their simplest forms. We can see an example of truth tables being shown to prove equivalence in HW2 section 1.2, exercise 8. The examples show that that based on different truth values for P and Q we receive the same values for two equations written differently, $\neg P \vee Q$ and $(Q \wedge P) \vee \neg P$. Since we know these equations are similar we can the create laws that can be used to manipulate expressions to simpler forms. In this case $(Q \wedge P) \vee \neg P$ can use the distributive law and tautology laws to simplify the statement to $\neg P \vee Q$.

Only a portion of the problem is included in order to save save space.

Problem 1. *How To Prove It: A Structured Approach*; Section 1.2, Solution to Exercise 8 : Use truth tables to determine which of the following formulas are equivalent:

Proof. (b) Truth Table for $\neg P \vee Q$

P	Q	$\neg P \vee Q$
F	F	T
F	T	T
T	F	F
T	T	T

P	Q	$P \vee \neg Q$	$Q \vee \neg P$	$(P \vee \neg Q) \wedge (Q \vee \neg P)$
F	F	T	T	T
F	T	F	T	F
T	F	T	F	F
T	T	T	T	T

(c) Truth table for $(P \vee \neg Q) \wedge (Q \vee \neg P)$

(e) Truth table for $(Q \wedge P) \vee \neg P$

P	Q	$Q \wedge P$	$(Q \wedge P) \vee \neg P$
F	F	F	T
F	T	F	T
T	F	F	F
T	T	T	T

Since the outputs are the same for both statements b) and e), b) and e) are equivalent, or:

$$\neg P \vee Q \equiv (Q \wedge P) \vee \neg P$$

□

Translating Sentences into Symbolic Form

Being able to translate sentences into symbolic for and vice versa is extremely important to being able to write and explain proofs and is the foundation to how we convey our ideas in mathematics. Mathematics should not be some cryptic language of symbols only mathematicians should be able to understand, rather we should be able to explain concepts in actual spoken word that can be deciphered by people who may not have the same level of math as you.

English sentences contain certain structures that allow us to group certain parts of sentences or words into mathematical statements and connectives that can be used to convey the same ideas in a more precise and efficient way. Translating sentences often includes using Letters like P, Q, R, etc or relations, written similarly to function notation that acts on certain variables in our statements. We then use these to represent key pieces of information in the statement and then we analyze the connectors that act on those parts of the statement. These connectors are words in the statement that relate one part of the statement to another. Common connectors like “or”, “and”, “implies”, etc, act on the different parts of the statement and change the conditions of what it means for the statement to be true or false.

Being able to translate sentences to and from symbolic form is extremely important when creating and understanding mathematical proofs. Whenever we are proving a theorem mathematically, we represent the theorem as a collection of statements with connectors and quantifiers to form a general argument that we validate. We can see a variety of examples of translating sentences to and from symbolic form down below.

§3.7 **Exercise 7**: To formally prove this statement, we will have to use the epsilon-delta definition of a limit. This definition states that no matter how much we zoom in or out of the point at which the limit is approaching, we should be able to find a distance from c, or δ, for every distance from $f(c)$, or ε. The logical statement is shown to be:

$$\forall\varepsilon > 0\exists\delta > 0\forall x(0 < |x - c| < \delta \rightarrow |f(x) - L| < \varepsilon)$$

In the write-up of our mathematical proof, we translate this logical definition into sentences using the context of the problem, using statements like "There exists a δ" and "which implies" in order to highlight the certain types of quantifiers and connectors and how we are using to them to validate our proofs.

Translating Definitions: Once we had reviewed the more basic proof techniques in this course, we used them to prove statements and theorems for different mathematical objects like sets, relations, and functions. All of the proofs from the artifacts section depend on interpreting how mathematical objects are defined and then we work to apply the correct proof methods from there. For example, the structure of existence and uniqueness proofs are based on interpreting the symbolic definition of Uniqueness.

For a unique x where $P(x)$ is true, or $\exists!x(P(x) : \exists x(P(x)) \wedge \forall y\forall z(P(y) \wedge P(z) \rightarrow y = z)$

We format this proof by proving there exists an x that makes $P(x)$ true ($\exists x(P(x))$), and then proceed to show that x is a unique solution $\forall y\forall z(P(y) \wedge P(z) \rightarrow y = z)$. Where the uniqueness portion of the proof states that for two values where P is true, then those values are the same.

Where the epsilon-delta and Uniqueness proofs are just two examples, translating sentences to and from symbolic form is the heart of mathematical proofs.

Direct Proof

Direct Proof is the method that we use to determine the validity of an argument through the process of deductive reasoning. When we use the method of direct proof we are addressing that whatever our hypothesis is, implies our conclusion. This gives us a useful way for us to determine the validity of the argument because we can follow that the logic of direct proof is that of a conditional statement. Given P as our premise or hypothesis, P implying Q, our conclusion, gives us the different ways we can interpret the argument based on its premises. This can be shown in the truth table for $P \rightarrow Q$.

Deductive reasoning is the process in which we use to prove an argument by direct proof. The process of deductive reasoning follows:

$$\begin{aligned} P &\to Q \\ P & \\ \hline \therefore Q & \end{aligned}$$

Where deductive reasoning is using information about P to deduce the conclusion, Q. As we assume P we can prove Q

In further analyzing the truth table for $P \to Q$, we can form syllogisms (logical process) that can be used when proving statements directly.

Modus Ponens: $P \to Q$ is true, P is true, thus Q must also be true

Modus Tollens: $P \to Q$ is false, Q is true, thus P must also be false

We can see the various examples of direct proof in the highlighted problems from the artifacts section:

- §3.7 Exercise 7: Proof shows that supposing $0 < |x-2| < \delta$, where $\delta = \frac{\varepsilon}{3}$, we can show $|f(x) - L| < \varepsilon$ for any $\varepsilon > 0$.
- §6.1 Exercise 4 and 9: Induction proofs follow the method of direct proof where $\forall n \in \mathbb{N}(P(n) \to P(n+1))$.
- §4.1 Exercise 5a) and 5b), §4.5 Exercise 10, §5.1 Exercise 20(b): Proving two sets are equivalent, is the same as proving a bi-conditional statement where being an element of one set implies its an element of the other and vice versa. This is used more formally as showing the two sets are subsets of each other.

Proof by Contradiction

Proof by contradiction is based on the idea that if we assume the negation, or opposite, of our premises, then we can determine information about the conclusion of the argument. Most importantly, when we do proof by contradiction, we are attempting to see when our premises are negated that we should come across a contradiction, or "mathematical nonsense" in our statements. Therefore, by coming across these contradictions and impossibilities, we have to say that our conclusion of the original statement must be true because of the problems we come across when its negation is assumed.

Most of the time, proof by contradiction is used as a "last resort" proof method, as mathematicians always aim to prove things directly and proof by contradiction doesn't give us any more logical information about an argument other than it has to be true or the laws of mathematics break down. Another reason contradiction proofs are controversial is that it can be difficult to find contradictions in the proof since there is no direction in the proof. Essentially, you are assuming the negation of the conclusion and hoping you find a contradiction.

We can see the various examples of proof by contradiction in the highlighted problems from the artifacts section:

- §3.2 Exercise 7: Assuming $a \notin C$ leads us to the contradiction $a \in B \setminus C$ which goes against the given $a \notin B \setminus C$. Thus, $a \in C$.
- §3.2 Exercise 10: if $x = 0$ then $y = 0$, which contradicts $y \neq 0$. Thus, we can conclude if $y \neq 0$, then $x \neq 0$.
- §3.6 Exercise 7: Contradiction is shown to use that any other statement than $A = B$, so $A \neq B$, leads to a contradiction. Thus $A = B$ is unique solution.

Proof by Contrapositive

Proof by contrapositive is similar to the method of direct proof whereas in a direct proof you are analyzing the argument $P \to Q$, in proof by contrapositive you are analyzing the statements contrapositive, which in this case is $\neg Q \vee \neg P$ which by definition is equivalent to $P \to Q$. In this method, we are essentially assuming $\neg Q$ in order to prove $\neg P$

Proof by contrapositive can be a very useful tool in order to understand different perspectives of mathematical arguments and proofs. This is because it is essentially equivalent to proving statements directly, but rather with the negation of the entire statement, which is accomplished using the contrapositive law for conditional statements.

There is only one proof for contrapositive in the artifacts section because the proof method is essentially an extension of direct proofs:

- §3.1 Exercise 8: originally proving $x \notin B$ implies $x \in B$. By proof by Contrapositive, we suppose $x \notin B$ and prove directly that $x \in D$.

I think the most difficult thing about proof by contrapositive is that I will accidentally stumble upon the proof method when trying to prove by contradiction. The most important thing to remember the difference between the two is that we are using the negation of the original conclusion in contrapositive proofs to find the negation of the original antecedent. Where in contradiction proofs we also assume the negation of the conclusion, but rather we use that to see if there is a contradiction with the original givens in our problem.

Proof Using Cases

Proof Using cases analyzes the logical forms of disjunctions in implications. In the most basic case, if we are trying to prove $P \vee Q$ implies R, where we are given P or Q are true, then we need to show that $P \to R$ and $Q \to R$.

$$
\begin{aligned}
&(P \vee Q) \to R \\
&(\neg P \wedge \neg Q) \vee R \ \textbf{Conditional Law} \\
&(\neg P \vee R) \wedge (\neg Q \vee R) \ \textbf{Distributive Law} \\
&(P \to R) \wedge (Q \to R) \ \textbf{Conditional Law}
\end{aligned}
$$

Where to prove the conjunction $(P \to R) \wedge (Q \to R)$, we prove $P \to R$ and $Q \to R$ as two separate proofs.

I think the most difficult aspect of proof by cases is not necessarily its application, but rather why splitting into cases works in the first place. We are not given this derivation in our book, but rather just told that this is how proofs in the form of disjunctions can be proved. It makes a lot more sense with the placement of §3.4 before §3.5 since proving disjunctions, by definition, means we have to know how to prove conjunctions in the first place.

We can see the various examples of proof by cases in the highlighted problems from the artifacts section:

- §3.5 Exercise 14: When asking to prove for every integer x, $x^2 + x$ is even, we need to consider the two possibilities for what x can be. Since x can be exclusively even or odd, we split the proof into two cases and prove from there.
- §4.1 Exercise 5a): When proving that$A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$, for our arbitrary element $(x, y) \in A \times (B \cup C)$, $y \in B \cup C$, we know $y \in B$ or $y \in C$, so we need to show for both cases $(x, y) \in (A \times B) \cup (A \times C)$. With the reverse implication, we follow a similar process but rather have $(x, y) \in (A \times B)$ or $(x, y) \in (A \times C)$.
- §4.5 Exercise 12: When proving statements for every n, we need to consider if n is even or odd, and show in both cases either $n^2 \equiv 0 \bmod(4)$ or $n^2 \equiv 1 \bmod(4)$.

Proof Using Existence and Uniqueness

As previously discussed, we use the logical definition $\exists x(P(x)) \wedge \forall y \forall z(P(y) \wedge P(z) \to y = z)$ to prove existence and uniqueness proofs. Since we have the conjunction joining $\exists x(P(x))$ and $\forall y \forall z(P(y) \wedge P(z) \to y = z)$ we format our proofs to prove existence and uniqueness separately.

The most challenging part of these proofs is proving the uniqueness of the solution. This process can seem rather abstract, as when you are proving uniqueness, you are proving that any two elements that satisfy your statement are ultimately equivalent. This is then tied back to the existence shown in the first part of the proof, where if this solution exists, then every other solution is equivalent to it. Therefore, the solution is unique. Ultimately, this can be confusing since we introduce many arbitrary variables in the second half of the proof, and the uniqueness proof can feel quite disjointed from the existence proof.

Existence and uniqueness proofs were, without a doubt, the most complex proofs for me to get down. The biggest thing that helped me understand these proofs was finding multiple ways to prove uniqueness. Finding a way to prove uniqueness through contradiction helped me draw connections to why the definition of why exactly $P(z)$ and $P(y)$ implies $y = z$. With practice and different approaches to proving uniqueness, I feel much more confident proving existence and uniqueness, as evidenced by the proofs in the artifacts section and the work that I have done with functions, which build heavily on this type of proof.

- §3.6 Exercise 9: Proving that the $A = B$ is the unique inverse operation of symmetric difference. Uniqueness is proved through contradiction.
- §5.3 Exercise 12: Proving the inverse function $f^{-1} : B' \to A$ shows that we use the definition of $f : A \to B'$ being onto shows that there exists a $b \in B'$ where $(b, a) \in f^{-1}$. Then uniqueness uses the fact that $f : A \to B'$ is one-to-one, where we can deduce that the two arbitrary variables a_1 and a_2 in A are equivalent.
- §5.3 Exercise 18: Existence and uniqueness are proved through theorems in the book, where to prove uniqueness, we can see that the composition of two functions is also a function with a unique output.

Proof By Induction

Proof by Induction is the most interesting proof technique we have learned so far in this course. Proof by induction is based on the idea that we can prove certain mathematical statements through an iterative process, which we used in this course to analyze the properties of the natural numbers. If we want to hypothesize a theorem about every element in a set, then the only way to prove this theorem holds is to test every element in the set. Not only is this process insanely long, but for infinite sets like the natural numbers, it would be impossible since there is an infinite amount of elements in the set of natural numbers. To get around this, we use a logical deduction that says for a supposed theorem about a set of elements, if we can prove that its next iteration is valid, then the supposed theorem is valid. Induction is proving that as long as we have a defined way to go to the next iteration in our formula or theorem, where supposing the theorem allows you to derive the next step. Then we should theoretically be able to implement this for every step in our proof, even if we are proving statements about a set with infinite elements.

To define the method of proof by induction on the natural numbers symbolically:

$\forall n \in \mathbb{N}(P(n) \to P(n+1)$

To rigorously prove these statements, we will begin with a base case where $n = 0$. If this case is not valid, then our theorem is not valid, and we cannot continue. Think of this step as the first rung on a ladder. If this step isn't valid, there's no way to get to the other steps to prove your theorem.

If the base case does in fact work out, then you can move on to supposing that your theorem is true and derive its next iteration through various methods involving algebraic manipulation, properties of even and odd numbers, definition of integer division, etc....

These proofs can be challenging at first and seem like outright magic at some points, but they all generally follow the same form, and once these are understood, induction proofs can be an extremely useful tool to prove properties of infinite sets.

We can see the various examples of proof by induction in the highlighted problems from the artifacts section:

- §6.1 Exercise 4: Proof by induction is used to prove a hypothesized theorem about the sum of odd numbers in a sequence. It is proved that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ since supposing this we know $1 + 3 + 5 + \cdots + (2n + 1) = (n + 1)^2$, which is the next iterative step of $n + 1$.
- §6.1 Exercise 13: Proof by induction is used in combination with integer division to show for all $n \in \mathbb{N}$ that $a - b \mid a^n - b^n$ since when supposing this we can prove $a - b \mid a^{n+1} - b^{n+1}$ which is the next step for $n + 1$.

Equivalence Relations

Equivalence relations are a specific type of equivalence relation defined by being reflexive, symmetric, and transitive.

Suppose R is an equivalence relation on A or in other words $R \subseteq A \times A$.

Reflexive: $\forall x \in A((x, x) \in R)$ or if the identity relation $i_A \subseteq R$.

Symmetric: $\forall x \in A \forall y \in A((x, y) \in R \to (y, x) \in R)$ or if $R = R^{-1}$.

Transitive: $\forall x \in A \forall y \in A \forall z \in A((x, y) \in R \wedge (y, z) \in R \to (x, z) \in R)$ or $R \circ R \subseteq R$.

We use equivalence relations to form mathematical objects called equivalence classes, where $[x]_R$ is the equivalence class, where for $y \in A$, y is related to x. An equivalence class

by definition allows us to group all elements with the same properties in the relation R which have the same properties. The book uses a good example of creating equivalence classes for vectors, where we can say that if two vectors have the same magnitude and direction, then they are the same vector. Each equivalence class would then be the set of all the vectors that point in the same direction with the same magnitude.

The book also covers partitions of Sets, which are defined as non-overlapping sets of subsets on a given set. If we have a partition on A defined by the equivalence relation R, or A/R, then we have disjoint sets of elements where $x \in A$ such that for a $y \in a,\ x \in [x]_R$.

Finally, we define the equivalence relation $x \equiv_m y$, or x is congruent to y modulo m, where we know $\exists k \in \mathbb{Z}(km = x - y)$. This is an equivalence relation that we often use to study properties of the natural numbers.

In my opinion, the most challenging part of equivalence relations was visualizing partitions. The way in which partitions are explained in the book makes them seem to be far more complex and abstract then they really are. It took a bit of outside research, but I found a video on partitions that uses Legos to explain partitions and sorts them into non-overlapping sets based on different properties. I thought this was a cool way of visualizing what seemed to be a complex mathematical operation to be something far simpler, allowing me to understand and formulate proofs using partitions.

We can see the various examples of equivalence relations in the highlighted problems from the artifacts section:

- §4.5 Exercise 10: Proof of that $\bigcup_{X \in \mathcal{F}}(X \times X) = R$. Where R is an equivalence relation on A and $\mathcal{F} = A/R$. Proof uses definitions of partitions effectively to prove that both sets are subsets of each other.
- §4.5 Exercise 12: Proof using definition of congruence to modulo m for even and odd integers.
- §5.1 Exercise 20a and 20b): Proof that the relation $R = \{(x, y) \in A \times A | g(x) = g(y)\}$, where $g : A \to B$, is an equivalence relation using definitions of reflexivity, symmetry, and transitivity. Then in part b) the emergence of equivalence classes are used to show that R and $\{(x, y) \in A \times A | g(x) = g(y)\}$ are equal where $g : A \to A/R$.

Functions

Functions are another example of relations with unique properties. Functions are relations from one set to another where each element of one set maps to a unique element in the other set. By definition for a function $f : A \to B$:

$$\forall a \in A \exists! b \in B((a, b) \in f)$$

We can also compose functions similar to relations given $f : A \to B$ and $g : B \to C$ then $g \circ f : A \to C$. We also know that composing functions is equivalent to nesting functions $g(f(a))$, which will be used in a majority of the function proofs in the artifacts section.

Given the definition of a function, we can go even further to classify functions as injective (one-to-one) and surjective (onto), where the following properties are met:

$$f : A \to B \text{ is injective} : \forall a_1 \in A \forall a_2 \in A(f(a_1) = f(a_2) \to a_1 = a_2)$$

$$f : A \to B \text{ is surjective} : \forall b \in B \exists a \in A(f(a) = b)$$

From these definitions, we can then go on to show that there exists an inverse function $f^{-1} : B \to A$, if f is bijective (both surjective and injective). This is also a result that is used for both proofs from §5.3 in the artifacts section.

The most challenging part of functions and mostly relations in general is getting mixed up in definitions and logical symbols. It can often be very helpful to create diagrams to visualize what functions and relations are actually doing and what exactly it means for a function to be one-to-one or onto.

We can see the various examples of equivalence relations in the highlighted problems from the artifacts section:

- §5.1 Exercise 20: Definitions of functions are used to analyze equivalence relations
- §5.2 Exercise 6: Proofs for $f : A \to B$ being one-to-one (injective) and onto (surjective),where $f(x) = ax + b$
- §5.3 Exercise 12: Proofs for such a $B' \subseteq B$ where $f^{-1} : B' \to A$ by definition of f being one-to-one and onto, modeled after theorem 5.3.1 in the book.
- §5.3 Exercise 18: Proof that there exists a function $h : A \to B$ such that $g \circ h = f$ where $f : A \to C$ and $g : B \to C$. Proof of useds theorems 5.1.5 and 5.3.1 to show the composition of two functions is a function and that bijectivity implies there exists an inverse of a function.

1.4 Actively Sharing Ideas

As mentioned, I am confident in my ability to actively share ideas and contribute to class discussion. In my opinion, we learn best when we aren't 100 percent correct from the start. Discussion very rarely should start with the right answer, but rather should be a continuous building of ideas until everyone achieves a full understanding of the material being taught. For example, when I present on the board, a lot of the time my work tends to be logically correct, but contains inconsistencies with the "mathematical grammar" that we use. By having people point out my mistakes, I am able to better reason my way through in order to display my thought process of why the mistake was made and then correct it for the class. Also, from my card contribution chart, It is evident that I continuously help those presenting in class by asking questions and assisting in presentations. I think I do a good job of going out of my way to give my input on questions that I might not be fully confident in, to give people who are presenting some sort of starting ground when they might get stuck.

1.5 Growth in Writing

In order to earn an A in this course, I have chosen to improve my mathematical writing. As mentioned, I have always struggled with being able to present my ideas in an efficient and cohesive manner, and I tend to include unnecessary information that is not useful to the explanation of the problem I am trying to solve. In homework assignment number 4 I discussed my goals for improvement in class and how I am able to show improvement. The goals and ideas listed were:

- Goal: Writing more concise, well-organized proofs that give the "right" amount of relevant explanation
- Goal: Get much more efficient in the amount of time it takes to explain, organize, and format mathematical ideas.
- Strengths: Most of the explanations given track my thought process and explain the problems and my solution fairly well.
- Weaknesses: Explanations tend to be pretty lengthy and go beyond the scope of what is needed. Writing up assignments in LaTeX tends to be a much lengthier process than what is needed.
- Showing improvement: periodic self-assessments for written-up assignments. What am I confident in? What am I not confident in? Is there anything I specifically thought I did well?

Since the start of this course, I believe that I have shown improvement in how I am able to explain my work and have made significant improvements in the time that it takes for me to write up and format assignments in LaTeX. Throughout the recent homework assignments, and even during the write-up of this portfolio, I have created a fairly consistent formatting in the explanation of homework problems, and by using certain built-in LaTeXfunctions, I

can type up my work much faster than I was in the start of the year. As someone who took the initiative to write up full LaTeXhomework problems before they were required, the time it takes me to complete homework has significantly decreased over the semester. Without having to spend multiple hours typing up homework, I have been able to put more time into actually working on writing proofs, which I have also shown significant growth in over the semester.

As highlighted in the last portfolio assignment, I would be focusing far more on the proof-writing aspect of this course than the write-up on LaTeX. I feel that throughout the second half of the semester, I have been able to communicate my thought process in proofs that follow in a well-organized manner that makes my proofs easy and efficient to read. I feel as if I've done a much better job with how I format my proofs from the start of my proof writing. I have gotten a much better grasp on when to implement certain proof techniques in the proof and how to transition different aspects of my proofs into one another. For example, when using definitions to manipulate anything in my proofs, I aim to follow the sort of U-shape that direct proofs are written out by, where I make sure to go through the important steps in which I've altered the givens and goals, and then wrap back around to prove the original statement.

One of the things I have done significantly better over the course of this semester is smoothing out the logical jumps in my proofs. The proofs that I have written towards the end of the course, I have focused on making sure each aspect of the proof leads directly into the next. This has made it much easier not to fall into the old habits I had formed when I first began writing proofs. From someone who constantly omitted statements about supposing that we have arbitrary elements and the important steps in my thought process, the proofs I wrote were very choppy and felt disjointed. They might have been logically correct, but they were not exactly written with the intent of logical progression. This is something I have definitely improved on over time by practicing with different proof techniques and proof structures. By no means are the proofs I am writing perfect, but I believe they are very acceptable for the level of this course and offer a good foundation for proof writing in higher-level courses.

All in all, I believe that I have respectfully accomplished the writing goals I have set for myself this semester and have shown improvement in the weaker aspects of my proof writing and LaTeXwrite-ups.

2 Reflection

2.1 Mid-Term Grade Reflection

a) What grade are you on track to earn?

As stated In this reflection, I am arguing for the grade of an A in this course and I believe that this is the grade that reflects my current understanding of course material, my participa-

tion in the classroom, my engagement with class discussion, and my growth in mathematical writing since the start of this course. As I have gone through and made my claims for the different aspects of earning an A throughout the other sections in the portfolio, I will focus more on the changes I have made to the style of learning I have, the mentality, and the heightened level of work that I put into this course.

b) How do you meet these requirements?

The grade requirements that are outlined in the syllabus in order to earn an A are:

- Understanding of all mathematical techniques
- Significant growth in mathematical writing or presenting
- Being a good Class Citizen
- Actively share ideas

I believe that my understanding of all the mathematical techniques learned in class so far is evident through the inclusion of explanations and artifacts surrounding each of the mathematical techniques in the previous section. I have included at least 1 and at most 7 examples for each of the topics and highlighted material towards the end of the course that contains a more cumulative representation of the material we have learned throughout this semester.

I have also shown significant growth in mathematical writing, specifically regarding proof writing. I have improved the flow of my proofs while providing an easy-to-follow and rigid proof structure for each proof technique discussed this semester. This, combined with the already presented growth in LaTeX, I am confident in my ability to communicate mathematical ideas and proofs in a more efficient and professional manner.

I think I have also done a good job of actively preparing for class in order to share my ideas and engage with class discussion, and that I am an all-around good class citizen. As mentioned, I have completed almost all of the work for this class on time and to the fullest ability. I am an active member of class discussion, and I do my best to make the classroom a supportive environment where people are allowed to make and learn from their mistakes. It is evident that I am not afraid to make mistakes, and that I am willing to present my work even if I am not 100 percent confident. I understand there is an importance to accepting that no one is perfect, especially in a learning environment, and that as long as I have an idea on how to approach a problem, class discussion can direct me towards a complete and correct solution, which contributes to the learning of the collective as a whole.

What's missing?

An important factor to take into account when determining my grade is the amount of effort that I have consistently put into my work and the changes that I have made in order

to be a better member of our class's community. As someone who is a perfectionist, it is very hard for me to go up to the board and present something that I know is not 100 percent correct. As I understand the importance of not being perfect on your first go around, this means that I have had to redefine my relationship with failure as a whole throughout this course, and that is something that has not easily been done. Regardless, there have been countless times in which I have stood up in front of the class for 20+ minutes, presenting something that not many other people were confident in, and I still am able to maintain my composure and eventually arrive at a complete solution. I think that not only have I been able to show I have improved in my mathematical writing, but in my presenting as well. As someone who isn't always the most confident presenting, I felt as if there have been several times where I have gone up to the board and delivered a well-explained explanation and solution to the problem that allowed people to ask questions and demonstrate their understanding of the material.

2.2 Short Answer

- 1. Finish the sentence with another student's name: "I especially appreciated ... because..."

  "I especially appreciated Rachel because she continuously does an excellent job of presenting in a neat and organized manner.

- 2. What was the most difficult part of the class for you so far? You can list a general topic, a specific problem, a type of work we've done, etc. Tell me why!

  I think the hardest part of this class for me was adjusting to the different course styles that we had. I am always so focused on the grades that I earn, and when it comes to tests and quizzes, it is very easy to know how well you are doing. However, with grades only updated twice a semester, there is a lot of uncertainty in how well you are doing in the course. I also really struggled with how subjective proof writing can be and adjusting to not always being given one specific way to approach certain problems.

- 3. What was the easiest part of the class for you so far? Tell me why!

  I don't know about easiest, but Induction proofs were fun. Especially the series ones, they felt more like puzzles, trying to get things to cancel into the next term.

- 4. What part of class has surprised or interested you the most so far? Why? One sentence!

  One of the subjects we touched on a lot was power sets, and I never really understood what they were used for, but after looking into cardinality and infinite sets, I realized they are a lot cooler than I thought they were.

3 Artifacts

Here is a collection of artifacts from coursework throughout this semester that highlights the understanding of the material taught in this class. I have already gone and explained the importance of each artifact in the mastery section of this paper. All of the proofs in this section connect to one or more proof techniques as my intended goal was to have at least 3 examples for each of the 10 topics.

Problem 2. *How To Prove It: A Structured Approach*; Section 3.1, Solution to Exercise 8 :

Suppose $A \setminus B \subseteq C \cap D$ and $x \in A$. Prove that if $x \notin D$, then $x \in B$.

Proof. Suppose $A \setminus B \subseteq C \cap D$ and $x \in A$. Assume $x \notin B$. Since $x \in A$ and $x \notin B$, then it is implied $x \in C$ and $x \in D$. Therefore if $x \notin B$, then $x \in D$. Since this is the contrapositive of the statement we began with, we can conclude that if $x \notin D$, then $x \in B$.

□

Problem 3. *How To Prove It: A Structured Approach*; Section 3.2, Solution to Exercise 7 :

Suppose that $A \subseteq B, a \in A$, and $a \notin B \backslash C$. Prove that $a \in C$

Proof. Suppose $A \subseteq B, a \in A$, and $a \notin B \setminus C$. Assume $a \notin C$. By definition of subset, $a \in A$ implies $a \in B$. Since $a \in B$ and $a \notin C$ then $a \in B \setminus C$. However, this contradicts the fact that $a \notin B \setminus C$. Therefore, $a \in C$.

□

Problem 4. *How To Prove It: A Structured Approach*; Section 3.2, Solution to Exercise 10 :

Suppose that x and y are real numbers. Prove that if $x^2y = 2x + y$, then if $y \neq 0$, then $x \neq 0$.

Proof. Suppose x and y are real numbers. Assume $x^2y = 2x + y$ and $y \neq 0$. Assume $x = 0$, then based on our equation, $y = 0$, which directly contradicts the fact that $y \neq 0$. Thus, we can conclude if $y \neq 0$, then $x \neq 0$.

□

Problem 5. *How To Prove It: A Structured Approach*; Section 3.5, Solution to Exercise 14 :

Prove that for every integer x, $x^2 + x$ is even.

Proof. (Case 1): Suppose x is an odd integer. Therefore, $x = 2k + 1$ where k is an integer, we know $x^2 + x = 4k^2 + 6k + 2$, which is even. Thus since k is an integer, $x^2 + x$ is even.

(Case 2): Suppose x is an even integer. Therefore, $x = 2k$ where k is an integer, we know $x^2 + x = 4k^2 + 2k$, which is even. Thus, since k is an integer, $x^2 + x$ is even.

Thus, for any $x \in \mathbb{Z}$, $x^2 + x$ is even. □

Problem 6. *How To Prove It: A Structured Approach*; Section 3.6, Solution to Exercise 9 (b):

Prove that every set has a unique inverse for the operation of symmetric difference. In other words, for every set A, there is a unique set B such that $A\Delta B = X$, where X is the identity element from part (a). (From part (a) we determined $X = \varnothing$ is the unique solution for $A\Delta X = A$.)

Proof. *Existence:* Suppose A, B and X are sets and $X = \varnothing$. Assuming $B = A$, then we know $A\Delta A = X = \varnothing$ and $B\Delta B = X = \varnothing$.

Uniqueness: Assume an arbitrary element $x \in A\Delta B$. For the sake of contradiction, assume $A \neq B$. Then, by definition of symmetric difference, $x \in (A \cup B) \backslash (A \cap B)$. By taking this difference, we are removing elements from $A \cup B$, where $x \in A$ and $x \in B$. Thus we are left with elements left where $x \in A$ or $x \in B$ exclusively. This contradicts the fact that $A\Delta B = X = \varnothing$ since $A\Delta B$ gives us a nonempty set when $A \neq B$. Thus since we arrive at a contradiction when $A \neq B$, then $B = A$ is the unique solution. □

Problem 7. *How To Prove It: A Structured Approach*; Section 3.7, Solution to Exercise 7:

Prove that $\lim_{x \to 2} \frac{3x^2-12}{x-2} = 12$.

Proof. Assume ε and x are arbitrary values, where $\varepsilon > 0$. Assume $\delta = \frac{\varepsilon}{3}$ and $0 < |x-2| < \delta$. With this value of δ, then we know $3|x-2| < 3\delta = 3(\frac{\varepsilon}{3})$. Simplifying, we get $3|x-2| < \varepsilon$, which, through algebraic manipulation, we can show is equivalent to $|\frac{3x^2-12}{x-2} - 12| < \varepsilon$.

$$
\begin{aligned}
|\frac{3x^2-12}{x-2} - 12| &< \varepsilon \\
|\frac{3(x^2-4)}{x-2} - 12| &< \varepsilon \\
|\frac{3(x-2)(x+2)}{x-2} - 12| &< \varepsilon \\
|3(x+2) - 12| &< \varepsilon \\
|3x+6-12| &< \varepsilon \\
|3x-6| &< \varepsilon \\
3|x-2| &< \varepsilon
\end{aligned}
$$

Thus, since we know there exists a δ for an arbitrary $\varepsilon > 0$ and $0 < |x-2| < \delta$ implies $|\frac{3x^2-12}{x-2} - 12| < \varepsilon$, then $\lim_{x \to 2} \frac{3x^2-12}{x-2} = 12$. □

Problem 8. *How To Prove It: A Structured Approach*; Section 6.1, Solution to Exercise 4: Find a formula for $1+3+5+\cdots+(2n-1)$, for $n \geq 1$, and prove that your formula is correct.

Examining the first four terms in this sequence, we can hypothesize that $1+3+5+\cdots+(2n-1)=n^2$ for $n \geq 1$. We can prove this with the following induction proof:

Proof. By Induction. Proving $1+3+5+\cdots+(2n-1)=n^2$, where n is an arbitrary element of $\mathbb{N}$.

Base Case: Suppose $n=1$, then $1=(1)^2$.

Inductive Step: Suppose $1+3+5+\cdots+(2n-1)=n^2$ for $n \geq 1$. Then it follows:

$$1+3+5+\cdots+(2n-1)=n^2 \tag{1}$$
$$1+3+5+\cdots+(2n-1)+(2n+1)=n^2+(2n+1) \tag{2}$$
$$1+3+5+\cdots+(2n+1)=n^2+2n+1 \tag{3}$$
$$1+3+5+\cdots+(2n+1)=(n+1)^2 \tag{4}$$

*From steps 1 to 2 we add $(2n+1)$ to both sides since it is the next term our sequence:
$$(2(n+1)-1)=(2n+1)$$

Since our inductive hypothesis holds for the case $n+1$, where $1+3+5+\cdots+(2n+1)=(n+1)^2$ then $1+3+5+\cdots+(2n-1)=n^2$ for $n \geq 1$.

□

Problem 9. *How To Prove It: A Structured Approach*; Section 6.1, Solution to Exercise 13:
Prove that for all integers a and b and all $n \in \mathbb{N}, (a-b) \mid (a^n-b^n)$.

Proof. By Induction, Suppose arbitrary $n \in \mathbb{N}$ and arbitrary $a, b \in \mathbb{Z}$.

Base Case: $n=0$, $(a-b)\cdot 0 = a^0-b^0 = 1-1=0$. Thus, $(a-b)|(a^0-b^0)$.

Inductive Step: Suppose $(a-b) \mid (a^n-b^n)$. Then we know an integer k exists, where $k(a-b)=a^n-b^n$. For the case of $n+1$, where $(a-b) \mid (a^{n+1}-b^{n+1})$ we can show:

$$\begin{aligned} a^{n+1}-b^{n+1} &= a(a^n-b^n)+a\cdot b^n-b\cdot b^n \\ \ldots &= a(k(a-b))+a\cdot b^n-b\cdot b^n \\ \ldots &= ka(a-b)+b^n(a-b) \\ a^{n+1}-b^{n+1} &= (ka+b^n)(a-b) \end{aligned}$$

Where, $(ka+b^n)\cdot(a-b)=a^{n+1}-b^{n+1}$, since $(ka+b^n) \in \mathbb{Z}$. Thus since it holds for $(a-b) \mid (a^{n+1}-b^{n+1})$, then for all integers a and b and all $n \in \mathbb{N}, (a-b) \mid (a^n-b^n)$.

□

Problem 10. *How To Prove It: A Structured Approach*; Section 4.1, Solution to Exercise 5:

Prove:
(a) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
(b) $(A \times B) \cap (A \times C) = (A \cap C) \times (B \cap D)$

(a) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

Proof. ($\rightarrow$) Suppose p is an arbitrary element of $A \times (B \cup C)$. Then by definition of the Cartesian product, p is an ordered pair (x, y) where $x \in A$ and $y \in B \cup C$. Since $y \in B \cup C$, then we can have $y \in B$ or $y \in C$.

Case 1: Suppose $x \in A$ and assume $y \in B$. Then we have an $(x, y) \in A \times B$. Which $A \times B$ is a set included in the union $(A \times B) \cup (A \times C)$. Thus $(x, y) \in (A \times B) \cup (A \times C)$.

Case 2: Suppose $x \in A$ and assume $y \in C$. Then we have an $(x, y) \in A \times C$. Which $A \times C$is a set included in the union $(A \times B) \cup (A \times C)$. Thus $(x, y) \in (A \times B) \cup (A \times C)$.

Therefore $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$.

($\leftarrow$) Suppose p is an arbitrary element of $(A \times B) \cup (A \times C)$. Then by definition of the Cartesian product, p is an ordered pair (x, y), where $(x, y) \in (A \times B) \cup (A \times C)$. Then by definition of the union of two sets, we know $x \in A$ and $y \in B$, or $x \in A$ and $y \in C$.

Case 1: Suppose $x \in A$ and assume $y \in B$. Since $y \in B$ is an subset of the union $B \cup C$ and $x \in A$, then $(x, y) \in A \times (B \cup C)$.

Case 2: Suppose $x \in A$ and assume $y \in C$. Since $y \in C$ is a subset of the union $B \cup C$ and $x \in A$, then $(x, y) \in A \times (B \cup C)$.

Therefore $(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$.

Thus, since $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$ and $(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$, $A \times (B \cup C) = (A \times B) \cup (A \times C)$

□

(b) $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$

Proof. ($\rightarrow$) Suppose p is an arbitrary element of $(A \times B) \cap (C \times D)$. By definition of the intersection of a Cartesian product, p is an ordered pair (x, y) where $(x, y) \in (A \times B)$ and $(x, y) \in (C \times D)$. Then we know $x \in A$ and $x \in C$, as well as $y \in B$ and $y \in D$. This can be rewritten as $x \in A \cap C$ and $y \in B \cap D$, so $(x, y) \in (x \in A \cap C) \times (y \in B \cap D)$. Thus,

$(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \cap D)$.

($\leftarrow$) Suppose p is an arbitrary element of $(A \cap C) \times (B \cap D)$. Then, by definition of the cartesian product, p is an ordered pair $(x, y) \in (A \cap C) \times (B \cap D)$, where $x \in (A \cap C)$ and $y \in B \cap D$. By the definition of the intersection of two sets, we know $x \in A$ and $x \in C$, as well as $y \in B$ and $y \in D$. Then we can say $(x, y) \in A \times B$ and $(x, y) \in C \times D$, so $(x, y) \in (A \times B) \cap (C \times D)$. Thus, $(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (C \times D)$.

Thus, since $(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \cap D)$ and $(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (C \times D)$, $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

□

Problem 11. *How To Prove It: A Structured Approach*; Section 4.5, Solution to Exercise 10:

Suppose R is an equivalence relation on A. Let F $= A/R$, and let S be the equivalence relation determined by $\mathcal{F}$. In other words, $S = \bigcup_{X \in F}(X \times X)$. Prove that $S = R$.

Proof. Suppose R is an equivalence relation on A and let $\mathcal{F} = A/R$. Suppose S is an equivalence relation determined by $\mathcal{F}$ where $S = \bigcup_{X \in F}(X \times X)$.

($S \subseteq R$)($\rightarrow$): Suppose an arbitrary $(x, y) \in S$. Then, since we know $S = \bigcup_{X \in F}(X \times X)$ where $\mathcal{F} = A/R$ or the partition of A into its equivalence classes on the relation R. Then we know $(x, y) \in X \times X$ which implies x and y are in the same equivalence class. By the definition of equivalence classes, then $(x, y) \in R$. Thus $S \subseteq R$.

($R \subseteq S$)($\leftarrow$): Suppose an arbitrary $(x, y) \in R$. Then we also know $x \in A$ and $y \in A$ where the equivalence relation R holds. Thus, we can say x and y are in the same equivalence class. Since $S = \bigcup_{X \in F}(X \times X)$ where $\mathcal{F} = A/R$, then we can say $(x, y) \in \bigcup_{X \in F}(X \times X)$, or $(x, y) \in S$. Thus, $R \subseteq S$.

Since, $R \subseteq S$ and $S \subseteq R$, $R = S$.

□

Problem 12. *How To Prove It: A Structured Approach*; Section 4.5, Solution to Exercise 12:

Prove that for every integer n, either $n^2 \equiv 0 \pmod 4$ or $n^2 \equiv 1 \pmod 4$.

Proof. Suppose we have an arbitrary $n \in \mathbb{Z}$. Since n can be either even or odd, we will examine both cases.

Case 1 (n is even): Suppose n is even. Then we know we can represent n as $n = 2k$ where k is an integer. Then we can say $n^2 = (2k)^2 = 4k^2$. We can then show $n^2 \equiv 0 \pmod 4$. By definition $n^2 \equiv 0 \pmod m$ can be written as $m \mid n^2 - x$ where there exists an integer i such that $m(i) = n^2 - x$. Taking $x = 0$ where $n^2 \equiv 0 \pmod 4$, we can rewrite as $4 \mid n^2$.

Since n is even, we can say $4 \mid 4k^2$. This is a valid case since there is an integer i, where $i = k^2$ such that $4i = 4k^2$. Thus, $n^2 \equiv 0 \pmod 4$ when n is even.

Case 2 (n is odd): Suppose n is odd. Then we know we can represent n as $n = 2k + 1$ where k is an integer. Then we can say $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$. We can then show $n^2 \equiv 1 \pmod 4$ for if n is odd. By definition $n^2 \equiv 0 \pmod m$ can be written as $m \mid n^2 - x$ where there exists an integer i such that $m(i) = n^2 - x$. Taking $x = 1$ where $n^2 \equiv 1 \pmod 4$, we can rewrite as, $4 \mid n^2 - 1$. Since n is even, we can say, $4 \mid 4k^2 + 4k + 1 - 1$ or $4 \mid 4k^2 + 4k$. This is a valid case since there is an integer i, where $i = k^2 + k$, such that $4i = 4k^2 + 4k$. Thus, $n^2 \equiv 1 \pmod 4$ when n is odd.

We can conclude, for n being even, $n^2 \equiv 0 \pmod 4$, and n being odd, $n^2 \equiv 1 \pmod 4$.

□

Problem 13. *How To Prove It: A Structured Approach*; Section 5.1, Solution to Exercise 20:

(a) Suppose $g : A \to B$ and let $R = \{(x, y) \in A \times A \mid g(x) = g(y)\}$. Show that R is an equivalence relation on A.

Proof. In order for R to be an equivalence relation on A, R must be reflexive, symmetric, and transitive.

Reflexive: Since R is defined as the set of all ordered pairs (x, y) in $A \times A$ where $g(x) = g(y)$. Then for any $x \in A$, $(x, x) \in R$ since $g(x) = g(x)$. Thus, R is reflexive.

Symmetric: Suppose $(x, y) \in R$. Then, since we know R is defined by $(x, y) \in A \times A$ where $g(x) = g(y)$, then since we know $g(y) = g(x)$, it must also be that $(y, x) \in R$. Thus, R is symmetric.

Transitive: Since R is defined as the set of all ordered pairs (x, y) in $A \times A$ where $g(x) = g(y)$, we can assume arbitrary x, y, and $z \in A$ as well as $(x, y) \in R$ and $(y, z) \in R$. Since $g(x) = g(y)$ and $g(y) = g(z)$, then it is also known $g(x) = g(z)$ and $(x, z) \in R$. Thus, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation on A.

□

(b) Suppose R is an equivalence relation on A and let $g : A \to A/R$ be the function defined by the formula $g(x) = [x]_R$. Show that $R = \{(x, y) \in A \times A \mid g(x) = g(y)\}$.

Proof. Suppose R is an equivalence relation on A and g is a function from A to A/R.

($\to$): Suppose an arbitrary $(x, y) \in R$. Since R is an equivalence relation on A, then $x \in A$ and $y \in A$. Since $[x]_R$ is defined as the set where $y \in A$ and $(x, y) \in R$, then we

also know by symmetry of R, $(y, x) \in R$, so $[x]_R = [y]_R$. Since $[x]_R = [y]_R$, then we know $g(x) = g(y)$. Thus $R \subseteq \{(x, y) \in A \times A \mid g(x) = g(y)\}$.

($\leftarrow$): Suppose an arbitrary $(x, y) \in \{(x, y) \in A \times A \mid g(x) = g(y)\}$. Since is an equivalence relation and g is governed by the form $g(x) = [x]_R$, we can say for $x \in A$ and $y \in A$, that $[x]_R = [y]_R$. Then we also know that since $[x]_R = [y]_R$, $(x, y) \in R$. Thus, $\{(x, y) \in A \times A \mid g(x) = g(y)\} \subseteq R$.

Since $\{(x, y) \in A \times A \mid g(x) = g(y)\} \subseteq R$ and $R \subseteq \{(x, y) \in A \times A \mid g(x) = g(y)\}$, $R = \{(x, y) \in A \times A \mid g(x) = g(y)\}$.

□

Problem 14. *How To Prove It: A Structured Approach*; Section 5.2, Solution to Exercise 6:

Suppose a and b are real numbers and $a > 0$. Define $f : \mathbb{R} \to \mathbb{R}$ by the formula $f(x) = ax + b$. Show that f is one-to-one and onto.

Proof. (*One-to-one*): Suppose arbitrary values x_1 and x_2 where $x_1 \in \mathbb{R}$ and $x_2 \in \mathbb{R}$. Assume $f(x_1) = f(x_2)$. Then we know $ax_1 + b = ax_2 + b$. Subtracting by b and dividing by a, since $a \neq 0$, we know $x_1 = x_2$. Thus, f is one-to-one.

(*Onto*): Suppose arbitrary $y \in \mathbb{R}$. Then if we assume $x = \frac{y-b}{a}$, we know $f(\frac{y-b}{a}) = a(\frac{y-b}{a}) - b$. Then we can simplify to $f(\frac{y-b}{a}) = y - b + b$ and then $f(\frac{y-b}{a}) = y$. Thus, since for an arbitrary $y \in \mathbb{R}$ where there exists an $x \in \mathbb{R}$ such that $f(x) = y$, then f is onto.

□

Problem 15. *How To Prove It: A Structured Approach*; Section 5.2, Solution to Exercise 10:

Suppose $f : A \to B$ and $g : B \to C$

(a) Prove that if $g \circ f$ is onto, then g is onto.

Proof. Suppose $f : A \to B$, $g : B \to C$, and $g \circ f$ is onto. Since $g \circ f$ is onto, then we know for all $c \in C$ there exists an $a \in A$ where $g \circ f(a) = c$, which can be written as $g(f(a)) = c$. Since $f : A \to B$, we can define $b = f(a)$. Then we know, $g(f(a)) = g(b) = c$. Thus g is onto.

□

(b) Prove that if $g \circ f$ is one-to-one then f is one-to-one.

Proof. Suppose $f : A \to B$, $g : B \to C$, and $g \circ f$ is one-to-one. Then for arbitrary values $a_1 \in A$ and $a_2 \in A$, we can assume $f(a_1) = f(a_2)$. Taking the composition $g \circ f(a_1) = g \circ f(a_2)$, where we know $g \circ f$ is one-to-one, then $a_1 = a_2$. Thus, f is one-to-one.

□

Problem 16. *How To Prove It: A Structured Approach*; Section 5.3, Solution to Exercise 12:

Suppose $f : A \to B$ and f is one to one. Prove there is some set $B' \subseteq B$ such that $f^{-1} : B' \to A$.

Proof. Suppose $f : A \to B$ and f is one-to-one. Assume B' is the set of all elements in the range of f, or $B' = Ran(f)$, where $B' \subseteq B$. Then we can proceed to prove $f^{-1} = B' \to A$.

Existence: By exclusion of elements $b \in B$ where $(a, b) \notin f$, then f would be onto for $f : A \to B'$ since for every $b \in B'$ there exists an $a \in A$ such that $f(a) = b$. Thus we know $(a, b) \in f$ so $(b, a) \in f^{-1}$.

Uniqueness: Suppose $(b, a_1) \in f^{-1}$ and $(b, a_2) \in f^{-1}$. Then we know $(a_1, b) \in f$ and $(a_2, b) \in f$. Since f is one-to-one, we know $f(a_1) = f(a_2)$ implies that $a_1 = a_2$.

Since there exists a unique $a \in A$ for all $b \in B'$, then $f^{-1} : B' \to A$.

□

Problem 17. *How To Prove It: A Structured Approach*; Section 5.3, Solution to Exercise 18:

Suppose $f : A \to C$, $g : B \to C$, and g is one-to-one and onto. Prove that there is a function $h : A \to B$ such that $g \circ h = f$.

Proof. Suppose $f : A \to C$, $g : B \to C$, and g is one-to-one and onto. Assume $h = g^{-1}(f(a))$. Since g is one-to-one and onto, we know by **Theorem 5.3.1** $g^{-1} : C \to B$. Then by the composition of two functions $g^{-1}(f(a)) : A \to B$, since $f : A \to C$, where $g \circ h = g(g^{-1}(f(a))) = f(a)$. Then by **Theorem 5.1.5** we know $h : A \to B$ since $h = g^{-1}(f(a))$, where the composition of two functions is also a function.

□